



MAPÚA MALAYAN COLLEGES MINDANAO

WeaveStream: Connecting Matigsalug Artisans to the Digital Marketplace

Presented By:

Lyka Sophia S. Jardeleza

Bachelor of Computer Engineering

2025

INTRODUCTION

Before I took the SS085 course, I honestly didn't think much about indigenous communities. I saw my path in computer engineering as something totally separate from their world, which I only knew from short documentaries and cultural displays in the media. This course completely changed my perspective. Understanding how indigenous peoples protect their lands and traditions despite modern challenges opened my eyes to their resilience. It made me see that the technical skills I'm learning aren't just for building apps for big companies. They can actually be powerful tools for social good. I realized I have a role to play in making sure technology helps rather than hinders, creating a bridge that allows indigenous rights and heritage to thrive in our digital age.

This new understanding is what sparked "WeaveStream." The project aims to tackle the economic isolation of the Matigsalug artisans by connecting their traditional crafts directly to the digital marketplace. Our main goal is simple but important: to help artisans gain both the confidence and tools to earn independently while preserving their cultural identity. We plan to do this by:

- Running hands-on workshops to teach practical digital skills like using a computer, navigating the internet, taking good product photos, and communicating with customers online.
- Working *with* the community to build a simple, easy-to-use online store that reflects their needs and culture.

- Helping them earn a fairer price for their work by cutting out the middlemen who currently take most of the profits.
- Creating a space online that doesn't just sell products, but also shares the important stories and traditions behind each woven textile and piece of beadwork.

As a Computer Engineering student, this project is the perfect way to apply what I've been studying. It's not just theory anymore. I'll be dealing with real hardware challenges, like setting up a reliable computer station in a remote area. I'll be designing and building a secure, user-friendly e-commerce website. I'll also be solving networking problems to ensure a stable internet connection. This hands-on experience shows how engineering can be a direct force for creating a more just and equitable society.

IDENTIFICATION AND DEFINITION OF THE PROBLEM

The Matigsalug artisans face a tough and unfair economic situation. Research shows that their incredible handmade goods are often sold for a fraction of their true value, which severely limits their potential for a sustainable livelihood [1]. The core of the problem is a broken system. Because they can't easily reach the customers who would appreciate their work, they have to rely on middlemen. These third-party buyers purchase the crafts for very low prices and then sell them in city markets or online for a big profit. This system traps artisans in low income and ignores the craftsmanship and culture embodied in their work.

Making things worse is a major digital gap. While more people in the community have smartphones these days, they often lack the specific know-how to use the internet for business. This digital divide is a well-known barrier holding back indigenous communities throughout the country [3]. Many artisans haven't learned how to take attractive photos of their work, write compelling descriptions, or manage sales and conversations with customers on an e-commerce site. Without these skills, they can't tell their own story or build their own brand, leaving them invisible in the vast online marketplace.

Addressing both issues at once offers a sustainable and meaningful solution. The solution isn't just to give them a website; it's to provide them with the knowledge to use it effectively. By giving the community both the technical training and their own dedicated platform, we can help them bypass the exploitative system entirely. This approach allows the Matigsalug to take full control of their economic future and share their cultural heritage with the world on their own terms.

PROPOSED SOLUTION

"WeaveStream" addresses these challenges with a straightforward, two-part plan built specifically for the Matigsalug community. First, we'll hold a series of practical, community-based workshops. We won't be teaching complex programming. Instead, we'll focus on immediately useful skills: how to turn on a computer, browse the web safely, use a phone's camera to make products look their best, and talk to customers online.

Second, and happening at the same time, we will develop a custom online marketplace designed for mobile phones. This isn't about just making a generic Facebook page for them. The platform will be built *alongside* community members, not just *for* them. It will have simple menus, use local Cebuano words, and, most importantly, include a "Story Behind the Craft" for every item. This helps customers understand and appreciate the cultural significance, adding real value. Unlike big, impersonal e-commerce sites, "WeaveStream" will be fully owned and managed by the community, allowing them to control their brand, share their narratives, and build direct relationships with their customers for the long term.

PLAN OF ACTION

The "WeaveStream" project will roll out in three key phases, making sure the community's voice guides us every step of the way.

- **Phase 1: Building Trust and Co-Designing (Months 1-2)**

The process begins with gaining trust and building genuine collaboration. The first and most important step is to seek Free, Prior, and Informed Consent (FPIC) from the Matigsalug tribal leaders, following the national guidelines to ensure the process is respectful and correct [2]. We'll then hold meetings and workshops to listen to their needs and ideas, co-designing the platform's features together. We'll also coordinate with the National Commission on Indigenous

Peoples (NCIP) and partner with local NGOs who already have strong, trusting relationships in the area.

- **Phase 2: Building and Training (Months 3-5)**

This phase has two tracks running at the same time. While a team of computer engineering students codes the online marketplace based on the community's feedback, another team will be in the community conducting the digital literacy workshops. We'll use donated laptops and a mobile internet hotspot to make it all work. To pay for the initial equipment, we'll look for funding from university grants, ask for support from the corporate social responsibility (CSR) programs of local tech companies, and maybe even start a small online crowdfunding campaign.

- **Phase 3: Launching and Sustaining (Months 6-ongoing)**

The final phase is all about launching the live "WeaveStream" site and making sure it keeps running long after we've left. We plan to mentor several youth and artisans so they can fully handle the platform themselves.—from adding new products to answering customer questions. To make it sustainable, a small part of the money from each sale will go into a community fund to pay for internet costs and any future maintenance. Knowing that internet can be unreliable, the platform will be designed to be "lightweight" and will have an offline mode, so they can update their product list even without a signal.

CONCLUSION

In short, the Matigsalug artisans are stuck. They have incredible talent, but they're trapped between a limited local market and a digital world that feels out of reach, which leads to them being taken advantage of financially. "WeaveStream" tackles this problem directly by pairing essential digital skills with a platform the community truly owns. This approach gives the weavers and artisans the power to earn what they're truly worth and to decide how their culture is presented and valued. This project helped me see that computer engineering can also protect culture and promote fairness—not just build technology.

REFERENCES

- [1] Macusi, E. S., Digal, L. N., Balaso, V. M., Andam, M. B., Bongcay, M. F., Gultiano, E. A., & Opiso, E. M. (2023). Indigenous knowledge systems and

practices (IKSPs), livelihood resources and aspirations of the Matigsalog and Ata tribes. *Sustainability*, 15(14), 11182. <https://doi.org/10.3390/su151411182>

- [2] National Commission on Indigenous Peoples. (2012). *The revised guidelines on free and prior informed consent (FPIC) and related processes* (Administrative Order No. 3, Series of 2012). https://legacy.doe.gov.ph/sites/default/files/pdf/e_ipo/02_ncips_procedures_and_requirements_for_energy_projects.pdf
- [3] OpenGov Asia. (2025, February 4). *The Philippines: Digital inclusion for indigenous communities*. <https://archive.opengovasia.com/2025/02/04/the-philippines-digital-inclusion-for-indigenous-communities/>